

Lipschitz dependence of the normalised chart integral on phase and amplitude

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Abstract

Unit 209 opens Astra #22's programme A2 (general- d stochastic transfer). Transferring Headline XIII to random phases needs remainder control along varying inputs; the mechanism is an eventual equi-Lipschitz estimate for the normalised chart integral $F_N(\xi, \eta) = I_N(\xi, \eta)/S_N$. Pointwise, $|e^{-\beta s^2 + \beta a s} - e^{-\beta s^2 + \beta a' s}| \leq \beta s |a - a'| e^{-\beta s^2 + \beta M s}$ for $|a|, |a'| \leq M$, $s \geq 0$; integrating,

$$|I_N(\xi, \eta) - I_N(\xi', \eta')| \leq \|\eta - \eta'\|_\infty G_0(N) + \beta A \|\xi - \xi'\|_\infty G_1(N),$$
$$G_0 = \int u^h e^{-\beta s^2 + \beta M s}, \quad G_1 = \int u^h s e^{-\beta s^2 + \beta M s},$$

with $s = Nu^k$ and $|\eta'| \leq A$. Both G_0/S_N and G_1/S_N converge (Headline XIII at constant phase M , exponents h and $h+k$; $G_1 = N \cdot I^{h+k}$), so there is $L = L(h, k, \beta, M, A)$ with $|F_N(\xi, \eta) - F_N(\xi', \eta')| \leq L(\|\xi - \xi'\|_\infty + \|\eta - \eta'\|_\infty)$ for all large N and all continuous inputs bounded by M and A on the cube (`normalised_lipschitz_eventually`). Review v11 (units 206–208) was clean; its documentation should-fixes are applied. Zero `sorry/axiom`; 9086 jobs.

1 The things formalised

```
theorem quadKernel_le_of_le (h) (hs : 0 ≤ s) (haM : a ≤ M) : quadKernel a s
  quadKernel M s
theorem quadKernel_sub_le (h) (hs) (ha : |a| ≤ M) (ha' : |a'| ≤ M) :
  |quadKernel a s - quadKernel a' s| ≤ s * |a - a'| * quadKernel M s
theorem chartIntegral_shift_mul : u in unitBox d, (u^h) * ((N u^k) *
  quadKernel M (N u^k))
  = N * chartIntegral d (phaseShift h k 1) k N (fun _ => M) (fun _ => 1)
theorem chartIntegral_sub_le (h) (hN : 0 ≤ N) (continuity) (| |, | ' | M) (| ' |
  A)
  (| - ' | ) (| - ' | on closedCube d) :
  |chartIntegral d h k N - chartIntegral d h k N ' ' |
    * chartIntegral d h k N (fun _ => M) (fun _ => 1)
    + * A * (N * chartIntegral d (phaseShift h k 1) k N (fun _ => M)
      (fun _ => 1))
noncomputable def leadScale (h k) (l N : ) : := N ^ (-(2 l)) * log N ^
  (multCount (ratioExp h k) l - 1)
theorem shifted_const_phase_tendsto : Tendsto (fun N => N * chartIntegral (d+1)
  (phaseShift h k 1) k N
  (fun _ => M) (fun _ => 1) / leadScale h k l N) atTop ( (phaseCoeff (phaseShift
  h k 1) k (l + 1/2) M 1))
theorem normalised_lipschitz_eventually (hk) (hl) (h) (hmin) (hatt) :
```

```

L, 0 L N in atTop, ' ', Continuous → ... → (bounds on
closedCube) →
|chartIntegral (d+1) h k N / leadScale h k l N
- chartIntegral (d+1) h k N ' ' / leadScale h k l N| L * ( + )

```

2 Mathematics

The decomposition $\eta K - \eta' K' = (\eta - \eta')K + \eta'(K - K')$, the monotonicity $K_a \leq K_M$ for $a \leq M$, and $|K_a - K_{a'}| \leq \beta s |a - a'| K_M$ (from $|e^x - e^y| \leq |x - y| e^{\max(x,y)}$, already in `MixedBlockEnvelope`) give the integrand bound; `norm_integral_le_of_norm_le` integrates it. The factor $s = Nu^k$ in G_1 is exactly the exponent shift $h \mapsto h + k$ (`prod_pow_phaseShift` with $j = 1$), so G_1/S_N converges by Headline XIII at $(h + k, \lambda + \frac{1}{2})$ after the same $N^{-2(\lambda+1/2)} = N^{-2\lambda}N^{-1}$ bookkeeping as in unit 200. The constant $L = (|C_0| + 1) + \beta|A|(|C_1| + 1)$ with C_0, C_1 the two limits is uniform in the inputs because G_0, G_1 do not depend on them.

3 Lean notes

- `rw [integral_const_mul]` unifies up to beta, so a constant written as `(fun _ => 1) u` still matches `?c * ?f`; when two constant-multiple integrals are present, give the constant explicitly (`integral_const_mul ( * A * )`) to steer which one is rewritten.
- `Real.rpow_neg_one` does exist on this pin (contrary to the `CLAUDE.md` note about `Real.rpow_neg_one` being missing) — it rewrote $N^{-1} = N^{-1}$ here without complaint. Keep the `CLAUDE.md` caveat in mind but test.
- Deriving $0 \leq A, 0 \leq \delta$ from the bounds at the point $0 \in [0, 1]^d$ avoids adding nonnegativity hypotheses.

4 Review v11 disposition

Statements of units 206–208 clean. Applied: unit 207’s docstrings now say “convergence at the selected normalisation; the limiting coefficient may vanish by cancellation”, state finite nonempty family and $m_a \geq 1$, and mention remainders and zero coefficients; unit 208’s docstring notes that partition-of-unity weights vanish and do not satisfy its strict positivity hypothesis as stated; unit 206’s docstring uses the actual theorem names and “not valid in general”.

5 Next

Unit 210: Lipschitz continuity of the limit functional $(\xi, \eta) \mapsto \text{phaseCoeff}$ in sup norm, and uniform convergence $F_N \rightarrow F$ on sup-norm-compact input sets (equi-Lipschitz + finite net + pointwise convergence). Unit 211: transfer to random inputs in $C([0, 1]^d)^2$.